Problem Application	GNN Methods	Problem Description	Resources
Text Classification	Graph ConvolutionalNetwork (GCN), Graph Attention Network (GAT)	A classic application of GNNs in NLPis Text Classification. GNNs utilize the inter-relations of documents or words to in fer document labels. GCN and GAT mod els are applied to solve this task. They con vert text to graph-of-words, and then use graph convolution operations to convolve the word graph. They show through exper iments that the graph-of-words representa tion of texts has the advantage of capturing non-consecutive and long-distance seman tics.	Yao et al., 2019, Survey onGNN for Text Classification
Neural MachineTranslation	GCN, Gated Graph Neu ral Network (GGNN)	The neural machine translation (NMT)is considered a sequence-to-sequence task. One of GNN's common applications is to incorporate semantic information into the NMT task. To do this, we utilize the Syn tactic GCN on syntax-aware NMT tasks. We can also use the GGNN in NMT. It converts the syntactic dependency graph into a new structure by turning the edges into additional nodes and thus edges labels can be represented as embeddings.	Bastings et al., 2017
Relation Extraction	Graph LSTM, GCN	Relation Extraction is the task of extract ing semantic relations from the text, which usually occur between two or more enti ties. Traditional systems treat this task as a pipeline of two separated tasks, i.e., named entity recognition (NER) and rela tion extraction, but new studies show that end-to-end modeling of entity and relation is important for high performance since re lations interact closely with entity informa tion.	Recent GNN-RE Survey
Image Classification	GCN, GGNN	Image classification is a basic computer vi sion task. Most of the models provide at tractive results when given a huge training set of labeled classes. The focus now is to wards getting these models to perform well on zero-shot and few-shot learning tasks. For that, GNN appears quite appealing. Knowledge graphs can provide the neces sary information to guide the ZSL (Zero shot learning) task.	Few-Shot Image Classifica tion, PMC

Object Detection /Interaction / Region Classification	Graph Attention Net work (GAT), GNN, Graph CNN, Graph LSTM	There are other applications of computervision tasks like object detection, interac tion detection, and region classification. In object detection, GNNs are used to calcu late RoI features; in interaction detection, GNN is message-passing tools between hu mans and objects; in region classification, GNNs perform reasoning on graphs that connect regions and classes.	Shi et al., CVPR 2020

Table 1: Summary of Applications of Graph Neural Networks with Methods, Descriptions, and Key Refer ences

Problem Applica tion	GNN Methods	Problem Description	Resources
Physics	Graph Neural Network,Graph Networks	Modeling real-world physical systems isone of the most basic aspects of under standing human intelligence. By repre senting objects as nodes and relations as edges, we can perform GNN-based reason ing about objects, relations, and physics in an effective way. Interaction networks can be trained to reason about the interactions of objects in a complex physical system. It can make predictions and inferences about various system properties in domains such as collision dynamics.	Battaglia et al., 2020
Molecular Finger prints	GCN	Molecular fingerprints are feature vectorsthat represent molecules. ML models pre dict the properties of a new molecule by learning from example molecules that use fixed-length fingerprints as inputs. GNNs can replace the traditional means that give a fixed encoding of the molecule to allow the generation of differentiable fingerprints adapted to the task for which they are re quired.	Recent Molecular GNNs

Protein InterfacePredictio n	GCN	This is a challenging problem with im portant applications in drug discovery. The proposed GCN-based method respec tively learns ligand and receptor protein residue representation and merges them for pairwise classification. At a molecular level, the edges can be the bonds between atoms in a molecule or interactions be tween amino-acid residues in a protein. On a large scale, graphs can represent inter actions between more complex structures such as proteins, mRNA, or metabolites.	PMC Protein Interfaces2024
Combinatorial Opti mization	GCN, GAT, GNN	Combinatorial optimization (CO) is atopic that consists of finding an optimal object from a finite set of objects. It is the base of many important applications in finance, logistics, energy, science, and hardware design. Most CO problems are formulated with graphs. In a recent work by DeepMind and Google, graph nets are used for two key subtasks involved in the MILP solver: joint variable assignment and bounding the objective value. Their neural network approach is faster than ex isting solvers on big datasets.	DeepMind MILP Solver2020
Graph Generation	GCN, GNN, LSTM,RNN, Relational-GCN	Generative models for real-world graphshave drawn significant attention for their important applications including modeling social interactions, discovering new chemi cal structures, and constructing knowledge graphs. The GNN based model learns node embeddings for each graph independently and matches them using attention mech anisms. This method offers good perfor mance compared to standard relaxation based techniques.	GraphRNN paper, Graph VAE paper

Table 2: Summary of Applications of Graph Neural Networks with Methods, Descriptions, and Key Refer ences